TPO1 – THE TERMINAL MEASURE AND PRESSURE ESTIMATES

Abstract. This focused outline develops the terminal measure and matter-pressure comparison in the QED3 construction. It states the controlled-flow and stopping hypotheses, the exact terminal identity for bounded Borel observables, the independent zero-sector Gaussian estimate, and the free and interacting partition estimates. General terminal coefficient bounds are separated from the reference-scale marked estimates used later for response.

Contents

1. Introduction and target statement 1
Proof overview 5
2. P. Imported photon, fermion and exact-step inputs 6
3. T01. The common tuned trajectory and stopping scales 8
4. T06. Zero flux equals a harmonic-averaged Gaussian law 9
5. T08. Terminal inverses, marked legs and coefficient norms 10
6. G. Gaussian domination in the zero-flux sector 13
7. T09. The photon tower and exact terminal measure 15
8. T10. Free pressure and mesh-uniform block-penalty twists 17
Appendix A. Detailed proof tasks and supporting statements 19

1. Introduction and target statement

TPO1 treats the terminal measure and pressure part of QEO1. It draws from Sections 21 and 22 of the master QED3 manuscript and retains identifiers T08–T11. The opening gives the common-pressure target under explicit imports and numerical hypotheses; T09 states the exact measure theorem independently.

Current proof manuscript: TPM1, The terminal measure and pressure estimates for lattice QED in three dimensions.

Overall plan: QEO1/T11. The overview boxes and statement-page return links are clickable.

Source editions. The 263-page master and its TeX source are the preserved comprehensive edition of QEM1. Its page, section and line locators below refer to that edition. Current companion manuscripts are cited by their codes and result names.

Original inputs: S1, curvature limits and the fourth cumulant; S2, chamber fermions and reflection positivity; S3, chamber renormalization and the fourth cumulant.

T11. The terminal remainder and common matter pressure

Model-specific. Overview Target.

Setting and imported hypotheses. Fix $0 < \rho < 1$, the construction constants of P, $0 < \bar{m} \le 1$, $e > 0$, and $0 < \varepsilon \le 1/200$. Assume the common-trajectory theorem T01, with all its admission hypotheses. Thus

$$h = L^{\lfloor \log_L(1/\bar{m}) \rfloor}, \quad \eta = L^{-N}, \quad K = N + \log_L h, \quad q = e\sqrt{\eta},$$
$$a_j = L^{j-N}, \quad e_j = e\sqrt{a_j}, \quad \mu_j = \bar{m} a_j, \quad v_j = -\log e_j, \quad r_j = \lceil v_j^r \rceil_L, \quad r \ge 2,$$
$$R_{\mathrm{top}} = hMr_K, \quad R^* = R_{\mathrm{top}}/L, \qquad \mathcal{T}(e) = \{hL^T : T \in \mathbb{Z}_{\ge 0},\ hL^T \le R_{\mathrm{top}}\}.$$

Here $L \ge 13$ is odd, $M \ge L^2$ is a power of L, and $\lceil\cdot\rceil_L$ denotes rounding up to a nonnegative integral power of L. For $R \in \mathcal{T}(e)$ and $N \ge N_{\min}(R,e)$, put $n = R/\eta$, $k = k_*(R) \ge 1$, $\ell_j = R/a_j$, and mark terminal values by a star. The threshold and stop are defined in T01. In particular

$$\ell_* = Mr_*, \qquad 0 < \mu_* \le 1, \qquad e_* \le e_K < 1, \qquad e_j^{\varepsilon} \le \tfrac{1}{4}.$$

Use one pair $(m_{0,N}(e), \epsilon^v_{0,N}(e))$ selected on R_{top} and its actual trajectory on each eligible smaller torus. Its bounds are

$$|m_j| \le \tfrac{1}{8} e_j^{3/4-8\varepsilon}, \quad \|\mathscr{E}_j\|_j^x \le 4e_j^{1/4-6\varepsilon}, \quad |\epsilon_j^v| \le 5c_{\mathrm{fl}} e_j^{1/4-6\varepsilon}.$$

The independent fields, local domains and weighted norms are specified in D. Assume the exact step and domain conclusions of P at every used level; the zero-flux identity T06; the finite projected-trace hypotheses of G; and the free-symbol hypotheses of T10. The unmarked terminal estimates in T08 apply with the actual positive μ_*. Its stronger reference-scale hypotheses are required only for the separately stated marked-operator conclusions.

Terminal numerical hypotheses. Fix an integer $p_0 \ge 1$, $p > 0$, and put

$$p_{0,j} = v_j^{p_0}, \qquad p_j = v_j^p, \qquad p_* = v_*^p, \qquad v_* \ge 8, \qquad p_0 v_*^{p_0-1} \ge 28.$$

Use the complete common calibration

$$c(\gamma) = \sum_{z\in\mathbb{Z}^3} e^{-\gamma|z|_\infty}, \qquad c_{\mathrm{bd}}(\gamma) = 3e^{3\gamma}c(\gamma),$$

$$C_0 \ge \max\{1, C_P^{\mathrm{ph}} c_{\mathrm{bd}}(\gamma_P), C^F, 4b_+ + 16C^F c(\gamma^F)\}, \qquad b_+ = 1,$$

$$b_- = 1 - L^{-1}, \qquad c_\kappa = \sup_{\square} \sum_{X \supset \square} e^{-\kappa d_M(X)} \le K_0.$$

Here the photon and fermion constants are those of P; c_{fl} is the flow constant of T01. Let C_Σ, γ_Σ be the block-twist constants in T10. At every stop occurring in the conclusion require

$$8C_0^2\ell_*^6(b_-^{-1} + \mu_*^{-1}) \le e_*^{-\varepsilon/2},$$
$$4(2C_0 + c_\kappa + 5c_{\mathrm{fl}} + 6)\ell_*^3 \le e_*^{-\varepsilon},$$
$$e_*^{1/4-9\varepsilon} \le \tfrac{1}{8}, \qquad 4C_0\ell_*^2 p_* < e_*^{-\varepsilon},$$
$$12\ell_*^3(e^{-p_{0,*}/2} + e^{-p_*^2/(2C_0^2)}) \le e_*, \qquad C_\Sigma e^{-\gamma_\Sigma \ell_*} \le e_*.$$

Also retain the common charge-ceiling condition $e_K^{1/4-9\varepsilon} \le 1/8$. These are the simultaneous hypotheses of this terminal theorem. Their realization by ordered constants is treated in QEO1/T16, outside this outline.

Normalization. The fine operator $D^{\mathrm{ch}}[\,\cdot\,]$ takes the absorbed bond phase as its argument. Thus an unabsorbed unit connection B occurs as $D^{\mathrm{ch}}[qB]$. Write $\boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e)$, with either sign allowed, and define

$$w_N(B) = e^{-\epsilon^v_{0,N} n^3} \frac{\det(D^{\mathrm{ch}}[qB] + \boldsymbol{\mu}_N)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)}.$$

Let $F_N = dC^1(n)$ carry its counting inner product and standard Gaussian γ_N, and let $A(F)$ be its coclosed mean-zero potential. For $\theta \in [-\pi,\pi]^3$, put

$$H_{\theta,\mu} = \frac{\theta_\mu}{qn}, \qquad d\nu^\theta(F) = w_N(A(F) + H_\theta)\, d\gamma_N(F), \qquad \bar{\nu} = \int \nu^\theta \frac{d\theta}{(2\pi)^3}.$$

The matter factor is

$$z_{r,n} = \int \frac{d\theta}{(2\pi)^3} \int d\gamma_N(F)\, \det(D^{\mathrm{ch}}[q(A(F) + H_\theta)] + \boldsymbol{\mu}_N)$$
$$= e^{\epsilon^v_{0,N} n^3} \det(D^{\mathrm{ch}}[0] + \bar{m}\eta)\bar{\nu}(1) > 0.$$

Only the free denominator uses the positive reference mass. The normalized harmonic-averaged curvature law is $\bar{\mu} = \bar{\nu}/\bar{\nu}(1)$. For the variant with any controlled real pair $(m_{\mathrm{in}}, \epsilon^v_{\mathrm{in}})$ from T01, use the same definitions with that pair in place of $(m_{0,N}(e), \epsilon^v_{0,N}(e))$. The common-pressure conclusion below uses the selected pair; the terminal measure and remainder also apply to this variant.

Target theorem. The exact downward coordinates and terminal Gaussian m_* of T09 identify the small-field event

$$\mathfrak{G} = \{|\zeta_j(\Upsilon)| \le p_{0,j} \ (j < k),\ |dA_*|_\infty \le p_*\}, \qquad \chi_*(A) = \mathbf{1}_{\{|dA|_\infty \le p_*\}}.$$

Write $c_{\theta,\mu} = \theta_\mu/(e_*\ell_*)$. The terminal fermion integral Y_* is defined in T09. Its exact measure identity at zero external connection gives

$$\nu^\theta(\mathfrak{G}) = \int d\mathsf{m}_*(A)\, \chi_*(A) Y_*(A + c_\theta).$$

Write $R^{\mathrm{tw}}_\theta = \det D_*(c_\theta)/\det D_*(0) > 0$. For real mean-zero A with $|dA|_\infty \le p_*$,

$$Y_*(A + c_\theta) = e^{-\epsilon^v_k \ell_*^3} R^{\mathrm{tw}}_\theta (1 + \mathfrak{r}_*(\theta, A)), \qquad |\mathfrak{r}_*| \le \epsilon_1 := \tfrac{1}{2} e_*^{1/4-9\varepsilon}.$$

Put

$$\epsilon_3 = 6\ell_*^3(e^{-p_{0,*}/2} + e^{-p_*^2/(2C_0^2)}), \qquad \epsilon_4 = 6\ell_*^3 e^{-p_*^2/(2C_0^2)}.$$

The positive measures satisfy

$$e^{-\epsilon^v_k \ell_*^3} R^{\mathrm{tw}}_\theta (1-\epsilon_1)(1-\epsilon_4) \le \nu^\theta(\mathfrak{G}) \le e^{-\epsilon^v_k \ell_*^3} R^{\mathrm{tw}}_\theta (1+\epsilon_1),$$
$$(1-\epsilon_3)\bar{\nu}(1) \le \bar{\nu}(\mathfrak{G}) \le \bar{\nu}(1).$$

Let p^0_{ch} be the free chamber pressure of T10 and set

$$p_F(N, e, \bar{m}) = p^0_{\mathrm{ch}}(\bar{m}\eta) + \epsilon^v_{0,N}(e), \qquad \epsilon_5(x) = C_6 \log \frac{1}{1 - e^{-\kappa_{\mathrm{strip}} x}} \quad (x > 0).$$

For every eligible R, N,

$$\left|\log z_{r,n} - p_F n^3\right| \le \epsilon_P(R) := \epsilon_5(\bar{m}R) + 3e_*^{1/4-9\varepsilon} \le \epsilon_5(1/L) + 3e_K^{1/4-9\varepsilon}.$$

In particular, for $n_* = R^*/\eta$ and $N \ge N_{\mathrm{ref}}(e)$, the same pair and pressure on R^* and $LR^* = R_{\mathrm{top}}$ give

$$\left|L^3 \log z_{r,n_*} - \log z_{r,Ln_*}\right| \le C_{\mathrm{mat}} := (L^3+1)(\epsilon_5(1/L) + 1).$$

The constant depends only on L, ρ and the fixed chamber weights. The bound is uniform in the cutoff, physical mass and charge in the admitted range. The common pair and pressure may depend on the cutoff and the fixed construction.

Proof work: the small terminal remainder. Gauge A to a representative A^g with $|A^g|_\infty \le 2\ell_*^2 p_*$. Photon row bounds give $|H^{\mathrm{ph}}_* A^g|_{\mathrm{each}} < e_*^{-\varepsilon}/2$. The disk

$$c_\theta + tH^{\mathrm{ph}}_* A^g \in \widetilde{R}_* \subset \widetilde{G}_*, \qquad |t| < 2\rho_*, \qquad \rho_* = e_*^{-3/4+4\varepsilon},$$

is therefore available. Coefficientwise Cauchy estimates with kernel $1/[t(t-1)]$, together with T08, bound the quadratic difference, mass term and activity sum by

$$|V_*|_{a_2} \le (2C_0 + c_\kappa + 4) e_*^{1/4-8\varepsilon} \ell_*^3, \qquad a_2 = C_0^{-1} e_*^{-1/4+\varepsilon}.$$

Normalize the terminal Gaussian by the one-sided substitution $\Psi \mapsto D_*(c_\theta)^{-1}\Psi$, leaving barred variables fixed. Its rows, including the identity on barred fields, fit inside weight a_2 by the first stop inequality. The second gives $|V'_*|_1 \le e_*^{1/4-9\varepsilon}/4 \le 1/32$. The unit-covariance Wick bound and $|e^{V'_*} - 1|_1 \le 2|V'_*|_1$ give ϵ_1.

Proof work: positivity, logarithms and common pressure. Apply the exact Borel identity with $\Phi = 1$. The remainder may be complex; the integral equals its real part because the fine measure and twist prefactor are positive. Use $\mathsf{m}_*(\chi_*) \geq 1 - \epsilon_4$ and the independent zero-sector complement bound $\bar{\mu}(\mathfrak{G}^c) \leq \epsilon_3$ to obtain the two-sided mass bounds before taking logarithms. The pointwise twist identity in T10 converts the coarse determinant ratio to the fine free determinant. Average in θ only after this comparison.

For any controlled pair, before bounding the terminal vacuum, the resulting pressure estimate is

$$\left|\log z_{r,n} - [p^0_{\mathrm{ch}}(\bar{m}\eta) + \epsilon^v_{\mathrm{in}}]n^3 + \epsilon^v_k \ell^3_*\right| \leq \epsilon_5(\bar{m}R) + 2e_*^{1/4-9\varepsilon}.$$

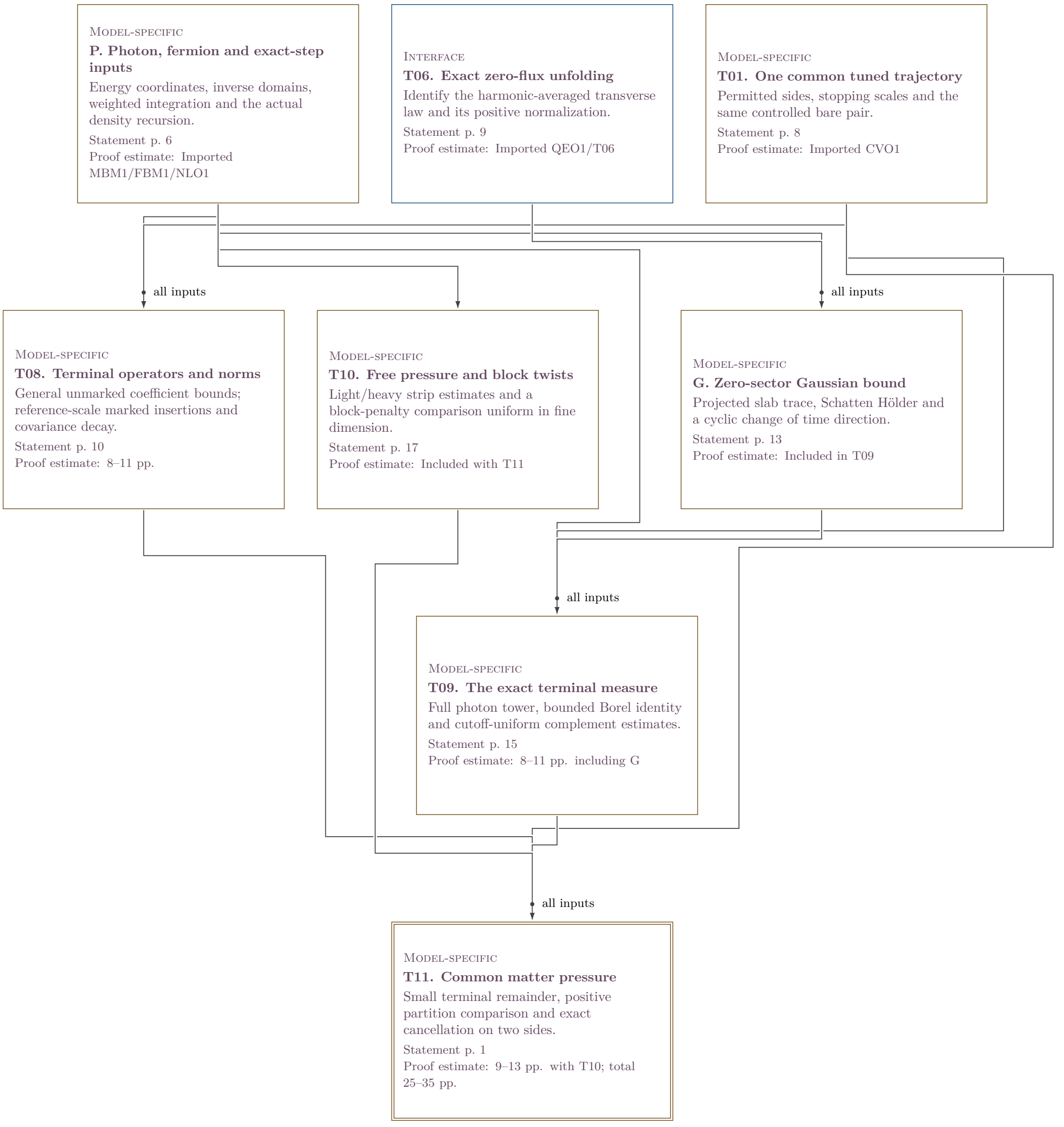
The common-pair vacuum bound in T01 and the second stop inequality bound $|\epsilon^v_k|\ell^3_*$ by $e_*^{1/4-9\varepsilon}$. Finally cancel p_F exactly between n^3_* and $(Ln_*)^3$, and use $\bar{m}R^* > 1/L$, $e_* \leq e_K$, and monotonicity of ϵ_5.

Source and scope. 263-page master Sections 21–22, especially `prop:term-remainder`, `prop:term-partition-bounds` and `thm:term-common-partition`, TeX 21572–21954. The exact Borel identity is `thm:term-density`, TeX 20865–20991. This is a focused outline of the supplied conditional terminal argument. Its full-proof allocation is 25–35 pages for T08–T11, excluding the imported primitive, tuning and zero-flux proofs.

TPO1 – The terminal measure and pressure estimates

Definitions/setup Model-specific Interface (general)

T11 states the complete pressure target; D gives the common activity definitions. P, T01 and T06 are imports. G states its additional projected-trace hypotheses. All numerical admission conditions remain in force.



Arrows record mathematical dependencies; no distinction between proof and application inputs is made here. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.

Click a box for its statement. Return links are navigation, not proof dependencies.

G supplies T09's complement bound. The tower and exact density have separate proofs. T08's marked clauses are retained for later response, which does not require the pressure conclusion. Estimates count each full-proof task once.

2. P. Imported photon, fermion and exact-step inputs

Model-specific. Overview Target.

One construction. Use the photon and chamber fermion primitives of MBO1/MBT and FBO1/FT, at the common constants and on the domains of D. Their full quantitative admission conditions are retained through NLO1/P and NLO1/S15. The maps at each step are the actual maps of that theorem. The coarse reference mass is positive; the independently tuned mass coupling is not incorporated into this reference inverse.

Photon coordinates and domains. At level j, $\ell_j = n/L^j$. Write $\mathbf{Q}$ for the unit block average, τ for the axial gauge map, and V_j for the Euclidean fluctuation-coordinate space. The linear maps $A_j^{\min}$ and Sq_j satisfy

$$\mathbf{Q}A_j^{\min}(A') = A'_L, \qquad \tau A_j^{\min}(A') = 0, \qquad \mathbf{Q}\mathrm{Sq}_j = \tau \mathrm{Sq}_j = 0.$$

The map $\mathrm{Sq}_j : V_j \to \ker \mathbf{Q} \cap \ker \tau$ is a bijection, and

$$\dim V_j = (2 - 2L^{-3})\ell_j^3, \qquad \|dH_j^{\mathrm{ph}}(A_j^{\min}(A') + \mathrm{Sq}_j\zeta)\|_2^2 = \|dH_{j+1}^{\mathrm{ph}}A'\|_2^2 + \|\zeta\|^2.$$

Here $H_0^{\mathrm{ph}} = I$, $\mathbf{Q}_j H_j^{\mathrm{ph}} = I$, and $d\mathbf{Q}_j = \mathbf{Q}_j^{(2)} d$. The plaquette average is a contraction for the indicated weighted Euclidean norms; the bond average preserves means. Its inverse scaling is $A'_L(x) = L^{-1/2}A'(x/L)$.

The real domain is $\mathrm{Dom}_j = \{A : H_j^{\mathrm{ph}} A \in \frac{1}{4}\widetilde{R}_j\}$, with arbitrary real harmonic part. A real field with $|dA|_\infty \le p_j$ belongs to it. Every $A' \in \mathrm{Dom}_{j+1}$ and $\zeta \in \mathrm{Box}_j = \{|\zeta(\Upsilon)| \le p_{0,j}\}$ give $A_j^{\min}(A') + \mathrm{Sq}_j\zeta \in \mathrm{Dom}_j$. These are the proved photon-domain inclusions, with all their charge and cutoff hypotheses imposed at every used level. The three reconstruction kernels have common bounds $C_P^{\mathrm{ph}} \exp(-\gamma_P d)$; their row sums enter C_0 in T08.

Fermions and weighted integration. The real chamber Dirac operator satisfies $\mathrm{Re}(D_A + \mu) \ge \mu I$ for $\mu > 0$. The fine-to-coarse average Q_A has analytic reverse $Q'_A = Q_{-A}^T$, with $Q'_A = Q_A^*$ and $Q_A Q_A^* = I$ for real fields. The normalized chamber-label measure is $1/48$; at fine mesh η_j, it is $\eta_j^3/48$. Barred and unbarred Grassmann generators are independent. Use the exact weighted Berezin, Wick and substitution formulas of NLO1/S07, including the $192n^3$ initial pairs and the four-component coarse spaces.

For the admitted inverse domain, the inverse S_*, field maps and short covariant quotients have the local kernel bounds stated in T08. Their calibration includes

$$0 < \gamma^F \le 1, \quad C^F \ge 1, \quad C^F \ge 608\exp(1)\max(1,b)a', \quad C^F \ge 64\exp(1)b^2a', \qquad b = 1,\ a' \ge 1.$$

The global all-pair component row bounds are also imported:

$$|H_j^{\#}(A)|_{1,\infty}, |\bar{H}_j^{\#}(A)|_{1,\infty} \le C^F, \qquad A \in \widetilde{G}_j.$$

They include distinct labels within one photon cluster and long allowed pairs. The three coefficient estimates in T08 use these row bounds for every $0 < \mu_j \le 1$.

For a real total lattice mass the two spin blocks are adjoints. Thus the fine determinant is nonnegative. Its zero-flux integral is strictly positive. This is the finite determinant input of QEO1/M01; it does not assert pointwise strict positivity at an arbitrary real mass. The free reference determinant at positive mass is strictly positive.

Exact integrated step. Along the full external-slot trajectory of T01,

$$y_{j+1}^x(A'; a^{[j+1]}) = \int_{\mathrm{Box}_j} d\mu_I(\zeta_j)\, y_j^x(A_j^{\min}(A') + \mathrm{Sq}_j\zeta_j; a^{[j]}).$$

The box restriction is unnormalized. The vacuum recursion already includes its probability cost. The initial activity and scalar external family are zero; the outgoing families satisfy $\mathscr{E}_j|_{C=0} = E_j$ and

$\mathfrak{P}_j(X,0)=0$. The exact finite density behind this identity is NLO1/S12; its nonlinear identification is NLO1/S15.

Use and source. These are imported mathematical conclusions, not new proof tasks. The zero-sector slab trace is specified separately in G; it is an additional finite-model input. 263-page master Sections 9–16, particularly the photon coordinate bijection and domain preservation, weighted block algebra, global propagator estimates and exact nonlinear density theorem. The same inputs appear explicitly at 263-page master TeX 19728–19905 and 20536–20570.

3. T01. The common tuned trajectory and stopping scales

Model-specific. Overview Target.

Imported result and physical parameters. Assume the theorem of CVO1/T01, including its actual-map and numerical hypotheses. Its proof remains in CVO1; we record precisely the conclusions used here. Fix $0 < \bar{m} \le 1$, odd $L \ge 13$, $M = L^m \ge L^2$, an integer $r \ge 2$, $0 < \varepsilon \le 1/200$, and physical charge $e > 0$. Put

$$h = L^{\lfloor \log_L(1/\bar{m}) \rfloor}, \quad \eta = L^{-N}, \quad K = N + \log_L h, \quad a_j = L^{j-N},$$

$$e_j = e\sqrt{a_j}, \quad \mu_j = \bar{m} a_j, \quad q = e\sqrt{\eta}, \quad v_j = -\log e_j, \quad r_j = \min\{L^i : i \ge 0 \text{ integral}, \ L^i \ge v_j^r\}.$$

The permitted sides are

$$R_{\mathrm{top}} = hMr_K, \qquad R^* = R_{\mathrm{top}}/L, \qquad \mathcal{T}(e) = \{hL^T : T \in \mathbb{Z}_{\ge 0}, \ hL^T \le R_{\mathrm{top}}\}.$$

At side R, write $n = R/\eta$, $\ell_j = R/a_j$, and let $k = k_*(R)$ be the first level with $\ell_j = Mr_j$. A star denotes the value at this stop. The terminal physical quantities and the permitted sides are independent of N.

Mesh threshold. For $R = hL^T$ and integers $s \ge 0$, set

$$\widehat{v}_s = v_K + \tfrac{s}{2}\log L, \qquad \widehat{r}_s = \lceil \widehat{v}_s^r \rceil_L, \qquad \widehat{n}_s = T + s - \log_L(M\widehat{r}_s).$$

At actual levels $s = K - j$. Here $\lceil u \rceil_L$ is the least nonnegative integral power of L at least u. If s_* is the largest zero of $\widehat{n}_s$, then

$$N_{\min}(R, e) = \max(0, s_* + 1 - \log_L h), \qquad k = K - s_* \ge 1.$$

These formulas and the existence of s_* are the stopping conclusions of CVO1/T01.a. For $N \ge N_{\min}(R, e)$, all used sides are admissible,

$$\ell_* = Mr_*, \quad Mv_*^r \le \ell_* < LMv_*^r, \quad 0 < \mu_* \le 1, \quad \mu_*^{-1} < L\ell_*, \quad e_* \le e_K,$$

and $\ell_j/(LM)$ is odd and at least 13. For $j < k$, also $\mu_j \le L^{-1}$ and $\ell_j \ge LMr_j$. Set $N_{\mathrm{ref}}(e) = N_{\min}(R^*, e)$. Only on R^* do we additionally use

$$a_* \in \{h/L, h/L^2\}, \qquad L^{-3} < \mu_* \le L^{-1}.$$

Common pair and controlled data. The bare pair $P_N(e) = (m_{0,N}(e), \epsilon^v_{0,N}(e))$ is chosen once on R_{top}. On every eligible smaller torus the same pair gives the actual trajectory of P, with

$$|m_j| \le \tfrac{1}{8} e_j^{3/4-8\varepsilon}, \qquad \|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le 4e_j^{1/4-6\varepsilon},$$
$$|\epsilon_j^v| \le 5c_{\mathrm{fl}} e_j^{1/4-6\varepsilon}.$$

The charge ceiling includes $e_j^\varepsilon \le 1/4$; hence the activity bound is within $w_j^E = e_j^{1/4-7\varepsilon}$. The no-slot and external-slot algebras and norms are those of D. The scalar couplings agree across the permitted tori up to their respective stops; the smaller terminal couplings need not vanish.

The pair is independent of R, the twist, external connection and probe within this fixed construction. It may depend on $N, e, \bar{m}$. The total fine mass $\bar{m}\eta + m_{0,N}(e)$ is real and has no prescribed sign. For estimates stated for any controlled real pair, use the same actual trajectory with $|m_j| \le e_j^{3/4-8\varepsilon}$ and $\|\mathscr{E}_j\|_j^x \le w_j^E$.

Use and source. The exact measure theorem uses a controlled trajectory on one torus. The last pressure comparison additionally uses this same selected pair on every side. 263-page master Section 16, Proposition 16.1 and Theorems 16.3–16.4; CVO1/T01 and its stopping and tuning proof tasks. No terminal partition lower bound enters the construction of the pair.

4. T06. Zero flux equals a harmonic-averaged Gaussian law

Interface. Overview Target.

Objects and hypotheses. On a rectangular cubical torus with sides $n_\mu \geq 4$, use counting pairings and $V = n_1 n_2 n_3$. Fix $q > 0$, $r_q = 2\pi/q$. Let $W(B)$ be a continuous weight on real bond cochains, invariant under real exact shifts and under r_q times integer cochains. Assume $W(B) \geq 0$ for real B and that its zero-flux integral, with the Maxwell Gaussian and compact-link measure, is finite and strictly positive. In the application $W(B) = \det(D_B + \mu_{\mathrm{lat}})$ with real total mass; determinant positivity and the nonzero $z = 0$ integral verify these hypotheses. For complex response assume W is entire and bounded on each imaginary strip uniformly in the real part. Let

$$V_g = dC^0, \quad V_h = \mathrm{span}\{\mathbf{1}_\mu\}, \quad V_t = (V_g + V_h)^\perp, \quad H_{\theta,\mu} = \theta_\mu/(qn_\mu), \quad \Delta_0 = d^*d.$$

For closed integer two-cochains, zero flux means zero sum on each coordinate two-torus. The photon curvature is $F = dA + r_q z$.

Claim. For exponentially bounded Borel Φ,

$$Z_n^{(0)}[\Phi] = \mathcal{N}_n \int_{[-\pi,\pi]^3} \frac{d\theta}{(2\pi)^3} \int_{V_t} \Phi(dA) e^{-\|dA\|^2/2} W(A + H_\theta)\, d\sigma(A),$$

where

$$\mathcal{N}_n = r_q^{V+2} (\det{}' \Delta_0)^{1/2}, \quad G_n = \frac{(2\pi)^{V-1}}{\det{}' \Delta_0}, \quad Z_n^{\mathrm{ph}} = \mathcal{N}_n G_n.$$

Here $Z_n^{(0)}$ sums zero-flux integer z and integrates each link of A over $[0, r_q)$. If γ_n is the standard Gaussian on $F_n = dC^1$ and $A(u)$ its transverse potential, then

$$Z_n^{(0)}[1] = Z_n^{\mathrm{ph}} z_{r,n}, \qquad z_{r,n} = \int \frac{d\theta}{(2\pi)^3} \int W(A(u) + H_\theta)\, d\gamma_n(u) > 0.$$

Normalizing gives equality of the two curvature probability laws.

Imported proof and normalization. Prove real Fourier Hodge decomposition and integer zero-flux exactness separately. Decompose closed integer one-cochains into an integer gradient plus three seam classes. Unfold the compact-link fundamental domain, retaining the coupling of exact and harmonic parts. Their Jacobians are $r_q^{V-1}(\det{}' \Delta_0/V)^{1/2}$ and $V^{1/2} q^{-3} d\theta$. On V_t, $\det(d^*d) = (\det{}' \Delta_0)^2$. These compute all normalization factors. Establish absolute convergence before Fubini, and integrable strip majorants before holomorphic parameter integration.

Inputs. Finite-dimensional gauge/phase invariance and strict zero-sector normalization. For the chamber determinant, prove its degree ≤ 32 in each oriented link phase and its imaginary-strip bound; the denominator free mass is not the total numerator mass.

Source. 263-page master Section 20, `eq:external-zero-sector`; QEO1/T06. The integer unfolding is imported here; its proof is not charged to TPO1.

5. T08. Terminal inverses, marked legs and coefficient norms

Model-specific. Overview Target.

Objects and hypotheses. Fine spinor blocks have total measure one per unit block, with four spin components and normalized chamber labels; coarse sites have four components and counting measure. Let Q_A be fine-to-coarse averaging and Q'_A its analytic transpose partner, equal to Q_A^* for real A. Assume $Q_A Q_A^* = I$ and $\mathrm{Re}(D_A + \mu_*) \geq \mu_* I$ for real A. Set

$$F_* = D_A + \mu_* + b_* Q'_A Q_A, \quad S_* = F_*^{-1}, \quad H_* = b_* S_* Q'_A, \quad D_* = b_* - b_* Q_A H_*.$$

General unmarked estimates. Assume $k_* \geq 1$, an admissible torus, $0 < \mu_* \leq 1$, $b_- = 1 - L^{-1} \leq b_* \leq b_+ = 1$, and the admitted charge ceiling and inverse domain of P. Its global all-pair row bounds apply throughout $\widetilde{G}_*$. For every real field, without a small-field condition,

$$D_*^{-1} = b_*^{-1} + Q_A (D_A + \mu_*)^{-1} Q_A^*, \qquad \|D_*^{-1}\|_2 \leq b_-^{-1} + \mu_*^{-1}.$$

Fix the calibration

$$c(\gamma) = \sum_{z \in \mathbb{Z}^3} e^{-\gamma |z|_\infty}, \qquad c_{\mathrm{bd}}(\gamma) = 3 e^{3\gamma} c(\gamma),$$

$$C_0 \geq \max\{1, C_P^{\mathrm{ph}} c_{\mathrm{bd}}(\gamma_P),\, C^F,\, 4 b_+ + 16 C^F c(\gamma^F)\}.$$

The positive constants $C_P^{\mathrm{ph}}, \gamma_P$ bound the photon reconstruction kernel by $C_P^{\mathrm{ph}} e^{-\gamma_P d}$. For $a_1 = e_*^{-1/4}$ and $a_2 = C_0^{-1} e_*^{-1/4+\varepsilon}$,

$$|F(H_*^{\#}\Psi)|_{a_2} \leq \|F\|_{h_*},$$

$$|\langle \bar{\psi}, \psi \rangle|_{a_2} \leq 4 e_*^{-1/2+2\varepsilon} \ell_*^3,$$

$$|\langle \bar{\Psi}, D_* \Psi \rangle|_{a_1} \leq C_0 e_*^{-1/2} \ell_*^3.$$

The coefficient norm $|\cdot|_a$ is the absolute coefficient sum weighted by a per generator. The fine kernel norm $\|\cdot\|_{h_*}$ uses its fine integration weights and $h_* = (e_*^{-1/4}, e_*^{-1/4+\varepsilon})$ for spinor and covariant-Hölder generators, respectively.

These are the estimates used by T11 on every permitted torus. In particular their constant does not use a positive lower bound on the running mass.

Reference-scale marked estimates. The following stronger conclusions are retained for the later response argument. In addition to the primitive hypotheses, assume $k_* \geq 1$, $\ell_* > 2M + 2$, $b_- = 1 - L^{-1} \leq b_* \leq 1$, $\mu_- = L^{-3} < \mu_* \leq L^{-1}$ and $\gamma^F M \geq 16$. On a symmetric open connection domain W, assume $e_* |\operatorname{Im} A| < 1/16$, $3M e_* |\operatorname{Im} A| < 1/4$, and the verified local inverse and short covariant-Hölder bounds $C^F e^{-\gamma^F d(y,y')}$ for S_*, H_*. Here $0 < \alpha < 1$, $\delta_{\alpha,A} v(x,x') = [U_A(x,x')v(x') - v(x)]/d_X(x,x')^\alpha$, Here U_A is the exponential of ie_* times the arithmetic mean of the six coordinate-order path sums, with the mesh factor, as in D. The short bound applies to $0 < d_X(x,x') < 1$, including distinct points within a chamber cluster, where the phase is one. Average phases use paths of length $\leq 3/2$; each path in a pair phase has length $\leq 3 d_X$. Allowed long pairs have $1 \leq d_X \leq M$. Distances between unit blocks are periodic sup distances. For a block-localized operator put

$$|O|(y,y') = \sup_{\|v\|_\infty \leq 1} \|\mathbf{1}_y O \mathbf{1}_{y'} v\|_\infty,$$

$$\|O\|_\gamma = \max\left\{ \sup_y \sum_{y'} |O|(y,y') e^{\gamma d(y,y')}, \sup_{y'} \sum_y |O|(y,y') e^{\gamma d(y,y')} \right\}.$$

Fine variables are localized in their unit blocks and pair variables at their first endpoint; the sup norm is the maximum of spinor component moduli. For a direction B define $\epsilon_B = e_* |B|_\infty$, P_B by supporting bond endpoints' unit blocks, and $P_B^+ = \{y : d(y, P_B) \leq M + 1\}$.

Claim. First derivatives of D_* are bounded by

$$226(C^F)^2\epsilon_B \sum_{z\in P_B} e^{-\gamma^F d(y,z)} e^{-\gamma^F d(z,y')}.$$

Put $\mathcal{S} = C^F(24/\gamma^F)^3$, $C_K = 226\mathcal{S}^2$, $C'_K = 1328\mathcal{S}^2$, $C_H = 203M(C^F)^2$. Writing $E(y,y') = e^{-\gamma^F d(y,y')}$, for $d(P_B, P_{B'}) \geq M+2$ the mixed bound is

$$|\partial_B \partial_{B'} D_*|(y,y') \leq C_H^2 \epsilon_B \epsilon_{B'} \sum_{\substack{z\in P_B \\ z'\in P_{B'}}} [E(y,z)E(z,z')E(z',y') + E(y,z')E(z',z)E(z,y')].$$

Let $\mathrm{Leg}_A(u;y')$ be any coefficient of a field $H_*^{\#}\Psi$ or its covariant quotient, barred or unbarred. Set $\mathrm{pos}(u)$ equal to its first endpoint's block, adding the second endpoint for long pairs, so it has at most two elements. The leg bounds are

$$|\mathrm{Leg}_A(u;y')| \leq 3C^F \sum_{w\in \mathrm{pos}(u)} E(w,y'),$$

$$|\partial_B \mathrm{Leg}_A(u;y')| \leq C_H \epsilon_B \sum_{\substack{w\in \mathrm{pos}(u) \\ z\in P_B^+}} E(w,z)E(z,y'),$$

$$|\partial_B \partial_{B'} \mathrm{Leg}_A(u;y')| \leq C_H^2 \epsilon_B \epsilon_{B'} \sum_{\substack{w\in \mathrm{pos}(u) \\ z\in P_B^+, z'\in P_{B'}^+}} [E(w,z)E(z,z')E(z',y') + E(w,z')E(z',z)E(z,y')].$$

The first derivative weighted row/column norm is at most $C_K\epsilon_B$ at rate $\gamma^F/2$, and its total block sum at most $C_K\epsilon_B|P_B|$.

At every real field, without a small-field assumption,

$$D_*^{-1} = b_*^{-1} + Q_A(D_A+\mu_*)^{-1}Q_A^*, \qquad \|D_*^{-1}\|_2 \leq b_-^{-1} + \mu_-^{-1}.$$

Its weighted component rows and columns are bounded by $\alpha_G = 2b_-^{-1} + 4\exp(1)\mu_-^{-1}(24/\theta)^3$, at rate $\vartheta_0 = \min(\theta/2, \gamma^F/8, 1/M)$, where $\theta = \mu_-/(12\exp(1))$.

The complex domains are explicit. Set

$$c_\tau = \min\{1/8, \mu_-/48, (292\mathcal{S})^{-1}\}, \qquad W^+ = \{A + A_1 : A\in W,\ e_*|A_1|_\infty \leq c_\tau\}.$$

On W^+, the inverse is analytic and

$$\|S_*\|_{\gamma^F/2} \leq 2\mathcal{S}, \qquad \|\partial_B D_*\|_{\gamma^F/2} \leq C'_K\epsilon_B, \qquad \sum_{y,y'} |\partial_B D_*|(y,y') \leq C'_K\epsilon_B|P_B|.$$

For $A = A_r + A_1$, with A_r real, $e_*|A_1|_\infty \leq \mu_*/24$ and $F_*(A)$ invertible,

$$D_*(A)^{-1} = b_*^{-1} + Q_A(D_A+\mu_*)^{-1}Q'_A, \qquad \|D_*(A)^{-1}\|_2 \leq g_1 := b_-^{-1} + 4/\mu_-.$$

Every differentiated path must remain in the stated domain.

Proof work: supported insertions and all-pair rows. Differentiate the local phases and $S_*F_* = I$, keeping the endpoint support of each insertion. For separated directions the two possible insertion orders give the two chains in the mixed estimate. A long covariant pair reads both endpoint blocks; bounding it as a short pair would lose one chain. Prove its row bound from the two ordinary field rows and its transport phase. Include distinct labels within a photon cluster, whose transport phase is one. Convert the localized bounds to global component row sums before substituting into the activity norm. The fine mass form has factor four because its 48 chamber labels are integrated with weight $1/48$.

Proof work: real and complex inverses. Accretivity and the finite-rank inverse identity give the real-field covariance at the actual μ_*. Conjugate the fine inverse by a periodic exponential distance weight with slope $\mu_*/(12\exp(1))$. Only the external hop changes; its perturbation is at most $\mu_*/2$. This yields exponential decay, followed by the component row and column bounds. On W^+, integrate the local derivative of F_*; the weighted perturbation times its inverse is at most $146c_\tau\mathcal{S} \le 1/2$. The Neumann series gives the enlarged inverse and derivative bounds. For the last complex covariance formula, the fine Dirac perturbation is at most $\mu_*/2$; the determinant identity for the finite-rank update supplies the remaining coarse invertibility. No complex continuation of a small-field real covariance bound is assumed.

Source. 263-page master Section 21.2–21.4, TeX 19728–20376. The general unmarked coefficient norms are stated in S3:54365–54373 and proved at 54214–54373. The marked insertion and complex-domain argument is S3:87137–88611; the retained real covariance proof is S3:43041–43387. Full-proof allocation: 8–11 pages. The reference mass interval belongs only to the marked and uniform-decay clauses.

6. G. Gaussian domination in the zero-flux sector

Model-specific. Overview Target.

Objects and hypotheses. Let $\bar{\mu}$ be the normalized harmonic-averaged zero-flux law with real lattice mass and coordinate sides ≥ 4. Let $F_N = dC^1$ have its counting inner product and standard Gaussian γ_N. The following coordinate-slice trace is an input, in each cyclic choice of time direction. For time direction 1, let Σ be the n_2 by n_3 slice, $r_q = 2\pi/q$, $v_s = n_2 n_3$, $\Gamma_s = r_q(\mathbb{Z}^{\Sigma^{(1)}} \cap \ker d_s)$, and $X_s = \mathbb{R}^{\Sigma^{(1)}}/\Gamma_s$. There are finite crossing spaces $\mathcal{F}', \mathcal{F}$, and

$$H^s = \bigoplus_{k\in\mathbb{Z}} L^2(X_s;\mathcal{F}'), \qquad H^l = \bigoplus_{k\in\mathbb{Z}} L^2(X_s;\mathcal{F}).$$

The index k is the slice magnetic flux; P_0 is its zero projection. A bounded r_q-periodic half-slab matrix $\widehat{N}(A) : \mathcal{F}' \to \mathcal{F}$ defines

$$(M_0\phi)_k(A) = e^{-\|d_sA + r_qk\mathbf{1}_{\sigma_0}\|^2/4}\widehat{N}(A)\phi_k(A),$$

where σ_0 is one fixed slice plaquette. Here F^{mag} is multiplication by $d_sA + r_qk\mathbf{1}_{\sigma_0}$, and F_j^{el} is the unfolded slice increment $A_{j+1} - A_j$. The commuting self-adjoint operators E_b generate translations $e^{-i\langle x,E\rangle}\phi(A) = \phi(A - x)$. If $u(a)$ is the crossing-space unitary with integer mode charges,

$$(P_G\phi)(A) = r_q^{-v_s}\int_{[0,r_q)^\Sigma} u(a)\phi(A - d_sa)\,da$$

defines the orthogonal gauge-average operator. It commutes with functions of E. For the two slice cycles S_2, S_3 and $\xi \in [0,1]^2$ set $b_\xi = q(\xi_2\mathbf{1}_{S_2} + \xi_3\mathbf{1}_{S_3})$. For bounded magnetic factors u_j and sub-Gaussian electric factors v_j, let K_j^ξ be the periodized convolution with kernel $v_j(A' - A)e^{i\langle b_\xi, A'-A\rangle}\prod_b g_1(A'(b) - A(b))$, where $g_1(t) = (2\pi)^{-1/2}e^{-t^2/2}$. The precise trace input is

$$Z^{(0)}\left[\prod_j u_j(F_j^{\mathrm{mag}})v_j(F_j^{\mathrm{el}})\right] = (2\pi)^{n_1n_2n_3}\int_{[0,1]^2}\mathrm{tr}\left[P_0\prod_{j=n_1-1,\dots,0} r_q^{v_s}K_j^\xi P_G M_0 u_j(F^{\mathrm{mag}})M_0^*\right]d\xi.$$

For a real two-cochain β take $u_j = e^{-\langle\beta_j^{\mathrm{mag}},F^{\mathrm{mag}}\rangle - \|\beta_j^{\mathrm{mag}}\|^2/2}$ and $v_j = e^{-\langle\beta_j^{\mathrm{el}},F^{\mathrm{el}}\rangle - \|\beta_j^{\mathrm{el}}\|^2/2}$. The needed source factorization is

$$K_j^\xi = z_jU_je^{-|E-b_\xi|^2/2}, \quad U_j = e^{i\langle\beta_j^{\mathrm{el}},E\rangle}, \quad |z_j| = 1,$$
$$K_\xi = r_q^{v_s}e^{-|E-b_\xi|^2/2}P_G, \qquad Y_j = K_\xi^{1/2}M_0\min(u_j,\delta^{-1})^{1/2}.$$

Y_j is Hilbert–Schmidt uniformly in ξ for every $\delta > 0$; all factors preserve magnetic flux. Cyclicity changes the truncated trace into $\prod_j z_j\,\mathrm{tr}\prod_j U_jP_0Y_j(P_0Y_j)^*$. These trace and factorization identities are assumptions to verify from the slab representation, rather than an assumption of Gaussian domination.

Conclusion. The law is the harmonic-averaged probability measure identified by T06, with the finite chamber positivity of P. For every real two-cochain g,

$$\bar{\mu}(e^{\langle g,F\rangle}) \leq \exp(\|\Pr_{F_N} g\|^2/2).$$

In particular, a real row λ on F_N of Euclidean norm at most one has $\bar{\mu}(|\lambda(F)| > t) \leq 2e^{-t^2/2}$ for $t > 0$. No terminal partition estimate is a hypothesis.

Proof work: preserve the sector projection. Use T06 to identify the normalized harmonic law. In the displayed truncated trace, apply Schatten Hölder to the $2n_1$ Hilbert–Schmidt factors $P_0 Y_j$ and their adjoints; the unitary electric factors have operator norm one. Then apply Hölder in ξ. Each factor of the resulting geometric mean is the zero-sector functional with a single magnetic insertion repeated on every time slice and with all electric insertions removed. The sector projector is retained in every factor. Let δ decrease to zero by monotone convergence. Choose a cyclic time direction in which the repeated magnetic plaquette is electric. The same inequality now bounds that factor by $Z^{(0)}[1]$. This gives the source inequality in the zero sector. Since $F \in F_N$, replace g by its orthogonal projection and complete the scalar square to obtain the displayed bound and its one-dimensional tails.

Standard trace input. For Schatten exponents with $\sum_i 1/p_i = 1$, $|\operatorname{tr} \prod_i A_i| \le \prod_i \|A_i\|_{\mathfrak{S}_{p_i}}$. The Hilbert–Schmidt hypothesis above implies membership in $\mathfrak{S}_{2n_1}$; multiplication by the projectors and unitaries is bounded. See Raphaël Ponge, *Singular Values and Schatten Classes*, Summer 2025, slides 3–9, 16 and 54–56 (`PongeSchatten2025`). The slab trace and its cyclic-coordinate applicability are separate model inputs; the trace inequality alone supplies neither.

Source. 263-page master Section 21.5, TeX 20377–20525, `eq:interacting-zero-flux-Bloch` and `prop:interacting-zero-flux-domination`. The source proof is S2:41900–42024,42387–42506 with the cyclic argument at 15730–15774; S3:83125–83229 gives the current dictionary. The all-flux estimate supplies integrability, not domination of the conditional law. G shares the 8–11-page full-proof allocation of T09.

7. T09. The photon tower and exact terminal measure

Model-specific. Overview Target.

Inputs and coordinates. Use the actual controlled trajectory of T01, the photon coordinates and exact integrated recursion of P, and the normalization $w_N, \nu^\theta, \bar{\mu}$ in T11. For a complex external fine connection a, define

$$d\nu^{\theta,a}(F) = w_N(A(F) + H_\theta + a)\, d\gamma_N(F).$$

This is a finite complex measure for complex a, and $\nu^{\theta,0} = \nu^\theta$. At stop $k \geq 1$, define

$$A_k^\downarrow = A, \qquad A_j^\downarrow = A_j^{\min}(A_{j+1}^\downarrow) + \mathrm{Sq}_j \zeta_j \quad (j < k).$$

Take A coclosed and mean zero on the terminal unit torus. Impose the axial conditions on all intermediate averages of the fine potential. These conditions define a slice on which d is a bijection onto $F_N = dC^1(n)$. It has dimension $2n^3 - 2$.

Gaussian coordinate theorem. The curvature has unique coordinates $(A_*; \zeta_0, \dots, \zeta_{k-1})$, with

$$\|F\|^2 = \|dH_k^{\mathrm{ph}} A_*\|_2^2 + \sum_{j<k} \|\zeta_j\|^2, \qquad \gamma_N \longmapsto d\mathsf{m}_*(A_*) \prod_{j<k} d\mu_I(\zeta_j).$$

Here m_* is the probability Gaussian on the terminal coclosed mean-zero bond space with density proportional to $\exp(-\|dH_k^{\mathrm{ph}} A\|_2^2/2)$. The coordinate map has Jacobian one for the volume measures of these energy scalar products. Fluctuation coordinates are orthonormal rows on F_N; each terminal plaquette row has norm at most one. Their counts are

$$\dim V_j = (2 - 2L^{-3})\ell_j^3 \leq 2\ell_*^3 L^{3(k-j)}, \qquad \#\{\text{terminal plaquettes}\} = 3\ell_*^3.$$

Closed coarse shifts descend to closed fine shifts. Their means scale by $L^{-k/2}$, and coarse gradients descend to fine gradients. In particular,

$$L^{-k/2} c_\theta = H_\theta, \qquad c_{\theta,\mu} = \theta_\mu/(e_* \ell_*).$$

Proof work: the full tower. At each level use the affine bijection and energy splitting from P. Their iteration gives a coordinate bijection on the axial slice. To prove injectivity of its curvature map, apply the telescoping energy identity to a zero-curvature field: every fluctuation vanishes, and the terminal coclosed mean-zero field vanishes. The dimension agrees with $\dim F_N = 2n^3 - 2$, giving surjectivity. Polarize the energy identity to obtain the isometry and its normalized Gaussian disintegration. Use the cochain identity $d\mathbf{Q}_k = \mathbf{Q}_k^{(2)} d$ and contraction of plaquette averaging for the plaquette-row bound. Preservation of means and the same identity give the closed-shift assertions. These statements concern the whole tower, not just one block.

Terminal fermion density. For a complex external fine connection a, require $a^{[j]} \in \mathfrak{X}_j$ at every $0 \leq j \leq k$, where $a^{[j]}(b) = L^{j/2} a(L^j b)$. Put $\mathscr{C}_* = a^{[k]}$. For $A \in \mathrm{Dom}_k$, let $\mathscr{A} = H_k^{\mathrm{ph}} A$, $B = \mathscr{A} + \mathscr{C}_*$, and evaluate the independent legs at $H_k^\#(B)\Psi$ and their barred and covariant-pair partners. Define

$$Y_*^x(A; \mathscr{C}_*) = \det D_k(0)^{-1} \exp\left(-\epsilon_k^v \ell_*^3 + \sum_X \mathfrak{P}_k(X, \mathscr{C}_*) \right)$$
$$\times \int \mathrm{D}\Psi \, \exp\left(-\langle \bar{\Psi}, D_k(B)\Psi \rangle - m_k \langle \bar{\psi}, \psi \rangle + \sum_X \mathscr{E}_k(X; \mathscr{A}, \mathscr{C}_*, \psi, \chi) \right).$$

The terminal integral has $4\ell_*^3$ pairs of weight one. The inserted fine fields and kernels retain weight $\eta_k^3/48$, where $\eta_k = L^{-k}$. The denominator is the positive reference determinant. At zero external slot, $\mathfrak{P}_k(X, 0) = 0$, and write $Y_*(A) = Y_*^x(A; 0)$.

Exact density theorem. On the stated domains the unnormalized box integral is

$$Y_*^x(A;a^{[k]}) = \int \prod_{j<k} (\mathbf{1}_{\mathrm{Box}_j}\, d\mu_I(\zeta_j)) w_N(A_0^{\downarrow}(A,\zeta)+a).$$

Define

$$\mathfrak{G} = \{|\zeta_j(\Upsilon)| \le p_{0,j}\ (j<k),\ |dA_*|_\infty \le p_*\},$$
$$\chi_*(A) = \mathbf{1}_{\{|dA|_\infty \le p_*\}}.$$

For every bounded Borel coarse observable Φ,

$$\nu^{\theta,a}(1_{\mathfrak{G}}\Phi(A_*)) = \int d\mathsf{m}_*(A)\, \Phi(A)\chi_*(A) Y_*^x(A+c_\theta;a^{[k]}).$$

The density on the right, extended by zero off $\{\chi_*=1\}$, is bounded and Borel. Its uncut terminal factor is jointly continuous on this support. The cutoff need not be continuous.

Proof work: domains, normalization and Borel events. Downward induction with P's domain-preservation statement makes every integrand legitimate. Iterate the exact density recursion, keeping the box measures unnormalized: their probability costs already appear in the vacuum recursion. The initial weighted determinant identity includes all $192n^3$ fine pairs and exactly the vacuum and denominator in w_N.

For a real fine field B_0, the chamber stencil gives the fixed-cutoff bound

$$|w_N(B_0+a)| \le \frac{e^{-\epsilon_{\mathrm{in}}^v n^3}}{|\det(D^{\mathrm{ch}}[0]+\bar m\eta)|}\big(|\bar m\eta + m_{\mathrm{in}}| + 41 + 6e^{q|\operatorname{Im} a|_\infty}\big)^{192n^3}.$$

This finite majorant justifies Fubini and the bounded Borel identity at each cutoff. It is not a uniform ultraviolet bound. Fine gauge invariance and the closed-shift identities replace the axial fine representative by $A(F)+H_\theta$. Dominated convergence on the compact boxes gives continuity of the uncut density on its stated domain.

Complement theorem. For $v_j = -\log e_j = v_* + (k-j)\log L/2$, put $p_{0,j} = v_j^{p_0}$ and $p_* = v_*^p$. Assume integer $p_0 \ge 1$, $v_* \ge 8$, $p_0 v_*^{p_0-1} \ge 28$, and $L \ge 5$. The independent zero-sector estimate G gives

$$\bar\mu(\mathfrak{G}^c) \le \epsilon_3 := 6\ell_*^3\left(e^{-p_{0,*}/2} + e^{-p_*^2/(2C_0^2)}\right),$$
$$\mathsf{m}_*(\chi_*=0) \le \epsilon_4 := 6\ell_*^3 e^{-p_*^2/(2C_0^2)}.$$

The first estimate is for the harmonic-averaged normalized law. No separate fixed-twist complement estimate is asserted.

Proof work: uniformity in depth. Apply the one-dimensional Gaussian tail to the actual fluctuation and plaquette rows. The fluctuation union bound contains

$$\sum_{i\ge 1} L^{3i} \exp\left[-\tfrac12 (v_* + i\log L/2)^{2p_0}\right].$$

The binomial lower bound using $p_0 v_*^{p_0-1} \ge 28$ leaves a summable factor L^{-4i} after the coordinate count. Thus the number of ultraviolet levels does not enter ϵ_3. The terminal plaquette union bound gives the second term; the same row bound under m_* gives ϵ_4.

Source. 263-page master Section 22.1–22.3, TeX 20536–21081: `prop:term-photon-tower`, `thm:term-density`, `prop:term-complement`. The photon tower is supplied in S1:19592–19610,19763–19926,20557–20754; the current density argument is S3:76861–77834 and the complement argument S3:83399–83505. Full-proof allocation: 8–11 pages for G and T09 together.

8. T10. Free pressure and mesh-uniform block-penalty twists

Model-specific. Overview Target.

Objects and hypotheses. For the 192-dimensional chamber symbol let $d_m(p) = \det D_m(p) > 0$ and $p^0_{\mathrm{ch}}(m) = \int \log d_m(p)\, dp/(2\pi)^3$. Use the periodic chamber symbol and its light–heavy decomposition from FBO1/S01, with four light and 188 heavy components: $D_m = \left(\begin{smallmatrix} D_{\mathrm{ll}} & D_{\mathrm{lh}} \\ D_{\mathrm{hl}} & D_{\mathrm{hh}} \end{smallmatrix}\right)$, with $B_m = D_{\mathrm{ll}} - D_{\mathrm{lh}} D_{\mathrm{hh}}^{-1} D_{\mathrm{hl}}$. For positive constants $g_{\mathrm{h}}, \theta_0, \kappa_{(0)}$ depending only on ρ, assume the following explicit strip bounds:

$$\|D_{\mathrm{hh}}(z)^{-1}\| \le 2/g_{\mathrm{h}} \quad (|\operatorname{Im} z_\nu| \le \theta_0),$$

$$\|\partial^\alpha B_m(p + ip'')^{-1}\| \le C_{\alpha,\rho}(|p|^2 + m^2)^{-(1+|\alpha|)/2} \quad \left(\begin{array}{c} |\alpha| \le 3,\ p \in [-\pi,\pi]^3, \\ |p''_\nu| \le \kappa_{(0)} m \end{array} \right),$$

$$\kappa_{(0)} m \le \theta_0/2, \qquad \|\partial^\beta B_m\| \le 24 K_s (2/\theta_0)^{|\beta|} \quad (1 \le |\beta| \le 4)$$

on the latter strip, with K_s independent of $m \in (0,1]$. The matrices and derivative traces are holomorphic on neighborhoods of the closed strips; their periodicity cancels the vertical contour sides. At zero mass the coercivity input is $\|D_0(p)u\| \ge c_{\mathrm{nz}}|p|_2\|u\|$.

For the block statement take odd $\mathfrak{N} \ge 3$, $\tau = \mathfrak{N}^{-1}$, a fine chamber torus of side ℓ with boundary multipliers $-e^{i\theta_\mu}$, and unit-block averaging Q with $QQ^* = 1$. For $0 < \mu \le 1$ and $0 < b_- \le b \le b_+$ define

$$T_\theta = D + \mu + bQ^*Q, \qquad D^{\mathrm{bl}}_\theta = b - b^2 Q T_\theta^{-1} Q^*.$$

The fine measure is $\tau^3/48$, and the coarse measure is counting.

Claim. There are $C_6, \kappa_{\mathrm{strip}} > 0$, depending only on the chamber parameter, such that

$$|\log\det(D^{\mathrm{ch}}[qH_\theta] + m) - n^3 p^0_{\mathrm{ch}}(m)| \le \epsilon_5(mn), \quad \epsilon_5(x) = C_6 \log(1 - e^{-\kappa_{\mathrm{strip}} x})^{-1}.$$

This is uniform in $n \ge 1$, $0 < m \le 1$ and real twist. Positivity gives the same bound for the logarithm of the harmonic-averaged determinant. At $m = \bar m \eta$ its argument is $\bar m R$. Moreover all three block determinants are positive,

$$(D^{\mathrm{bl}}_\theta)^{-1} = b^{-1} + Q(D + \mu)^{-1} Q^*, \quad \det D^{\mathrm{bl}}_\theta = b^{4\ell^3} \det(D + \mu)/\det T_\theta,$$

$$|\log(\det T_\theta/\det T_0)| \le C_\Sigma \ell^{-1} e^{-\gamma_\Sigma \ell},$$

with $C_\Sigma < \infty$, $\gamma_\Sigma > 0$ depending only on ρ, b_-, b_+, not on τ, μ, ℓ. At the terminal scale this identifies the coarse twist ratio with the fine free determinant ratio times $\det T_0/\det T_\theta$.

Proof work: free pressure. Differentiate the real logarithmic determinant and continue its derivative trace holomorphically to the stated strip. This avoids choosing a global branch of a complex logarithm. One integration by parts and a contour shift bound its Fourier coefficients. For the light block, retain the singular-order derivative estimates near $p = 0$; the heavy block has a uniform gap. Exact antiperiodic sampling selects the modes in $n\mathbb{Z}^3$. Summing them gives $\epsilon_5(mn)$, uniformly in twist. Use this pointwise positive estimate before harmonic averaging.

Proof work: the block inverse and fine dimension. In the Bloch cell, the nonzero chamber modes are controlled by $c_{\mathrm{nz}}|p|_2$, and the heavy modes by their gap. On the four low light modes, block averaging has size at least $8/\pi^3$; the block penalty therefore controls them even as μ decreases. Combine these estimates for a uniform real inverse, then enlarge to a fixed complex strip. In periodic gauge, for $1 \le j \le 4$,

$$\|\partial^j_\nu D(p)\| \le 9\tau^{j-1}, \qquad \|\partial^j_\nu (Q^*Q)(p)\| \le \exp(1), \qquad \operatorname{rank} \partial^j_\nu (Q^*Q)(p) \le 4(j+1).$$

Writing $G = T(p)^{-1}$, the supplied singular-value comparison and three-dimensional Bloch counting give

$$\tau^{4-s}\|G\|^s_{\mathfrak{S}_s} \le C_s, \qquad s = 1,2,3,4,$$

with constants independent of the fine dimension. For derivatives in one coordinate,

$$\partial^4 \log\det T = \operatorname{tr}\big(GT^{(4)} - 4GT'GT''' - 3GT''GT'' \\ + 12GT'GT'GT'' - 6GT'GT'GT'GT'\big).$$

Terms involving a derivative of Q^*Q use its finite rank. The other terms use the weighted Schatten bounds and the powers τ^{j-1}. The trace estimates are the standard ones recalled in G; the uniform singular-value comparison is part of this model-specific proof. Four integrations by parts, a strip shift and exact sampling give $C_\Sigma \ell^{-1} e^{-\gamma_\Sigma \ell}$. The heavy-cluster projection in the chamber decomposition and the block-average projection Q^*Q have different roles.

Terminal use. The finite-rank determinant identity gives the pointwise formula

$$R_\theta^{\mathrm{tw}} = \frac{\det(D^{\mathrm{ch}}[qH_\theta] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)} \frac{\det T_0}{\det T_\theta}.$$

The free comparison thus incurs $\epsilon_2 = C_\Sigma \ell_*^{-1} e^{-\gamma_\Sigma \ell_*}$. This step needs the chamber symbol estimates stated above, not a fourth-order continuum coefficient calculation.

Source. 263-page master Section 22.4, TeX 21082–21571, `prop:term-free-pressure` and `prop:term-block-twist`. The supplied proof is S3:43506–45225. Full-proof allocation: 9–13 pages for T10 and T11 together.

Appendix A. Detailed proof tasks and supporting statements

The activity definitions below fix the independent fields, weights, local domains and symmetry requirements imported by the terminal construction.

D. The activity space and its field domains

Setup. Overview Target.

Shared activity space. The terminal construction uses the same activity algebra as RNO1/D. We recall its complete norm, domains, support and generator actions here.

Chamber labels and evaluation. A chamber site is $x = (c, g)$, where $c \in \eta\mathbb{Z}^3/\ell\mathbb{Z}^3$, with $\eta = L^{-k}$ for a nonnegative integer k, and g ranges over the 48 signed permutation matrices in dimension three. There are four spin components and separate barred and unbarred generators. The three internal neighbours replace g by gs_i, where s_0 flips the third coordinate, s_1 exchanges the last two, and s_2 exchanges the first two. The external neighbour is $x^{(3)} = (c + \eta g e_1, g r_{e_1})$, where r_{e_1} flips the first coordinate. This neighbour convention fixes the external pairs used below. For a connection B, $U_B(x, x')$ is the exponential of ie_k times the arithmetic mean of its six coordinate-order path sums, each with the mesh factor η. It is one within a photon cluster. The pair generators remain independent until the evaluation

$$\chi(x, x') = \frac{U_{A+C}(x, x')\psi(x') - \psi(x)}{\max(|c - c'|_\infty, \eta)^\alpha}$$

and its barred version with inverse transport. Statements about an evaluated activity refer to this substitution.

The common setting. Define the activity space $\mathcal{A}_k^{x,\mathbb{C}}$, with real subspace $\mathcal{A}_k^x$. Fix odd $L \geq 5$, $\eta = L^{-k}$, $M = L^m \geq L^2$, $\ell/(LM)$ odd and at least 13, $0 < \varepsilon \leq 1/200$, $0 < \alpha < 1$, and $0 < e_k < 1$. An M-polymer is small if $d_M(X) \leq L$, with an injective covering lift. Put $\mathrm{Vol}(X) = M^3|X|_M$. Balanced means that each monomial has equally many barred and unbarred generators. An activity $E(X; A, C)$ is a balanced polynomial in independent ordinary and pair Grassmann legs, with holomorphic coefficients reading only bonds meeting X. Ordinary legs are over X; an allowed pair has all six coordinate-order paths in X, or is an external nearest-neighbor pair with an endpoint over X. Its separation is $d_X = \max(|c - c'|_\infty, \eta)$, and the pair range is $0 < d_X \leq M$; intracluster pairs are included. Index measures are $\eta^3/48$ and $(\eta^3/48)^2$, while generator weights are $h_1 = e_k^{-1/4}$, $h_2 = e_k^{-1/4+\varepsilon}$. Ordinary kernels use the antiperiodic convention in each chamber argument; pair indices are simultaneous period classes of their two endpoints. The coefficient norm of the separately antisymmetric kernels is

$$\|E(X; A, C)\|_{h_k} = \sum_{n,m} \frac{h_1^n h_2^m}{n!m!} \int |E_{nm}(X; A, C; \xi; \zeta)| \, \mathrm{d}\xi \, \mathrm{d}\zeta.$$

To specify its domains, put

$$\begin{aligned} \partial_\nu B_\mu(c) &= \eta^{-1}[B_\mu(c + \eta e_\nu) - B_\mu(c)], \\ \|B\|_1 &= \sup |B_\mu(c)|, \quad \|B\|_2 = \sup |\partial_\nu B_\mu(c)|, \\ \|B\|_3 &= \sup_{\mu,\nu} \sup_{0<d(c,c')\leq 1} \frac{|\partial_\nu B_\mu(c') - \partial_\nu B_\mu(c)|}{d(c, c')^\alpha}, \qquad \theta_i = e_k^{-3/4+i\varepsilon}. \end{aligned}$$

The distance is periodic sup distance; a local norm uses only its contained bonds and difference cells. $R_k = \{B : \|B\|_i < \theta_i,\ i = 1, 2, 3\}$. The lattice gradient is $d\lambda(c, c') = \eta^{-1}(\lambda(c') - \lambda(c))$. A field belongs to $\widetilde{R}_k$ if it is $A_0 + A_1$, A_0 real, $A_1 \in R_k$, and on every enlarged cube of side $(2L + 1)M$, $A_0 = A_\square + d\lambda_\square$ with real $A_\square, \lambda_\square$ and $\|A_\square\|_{i,\square\dagger} < \theta_i$. The smaller domain $\widetilde{R}_k^-$ retains the same local

real representatives and replaces the global bounds on A_1 by $\|A_1\|_i < (1-e_k^{\varepsilon})\theta_i$. The external domain is

$$\mathfrak{X}_k = \{C : \|C\|_1 < e_k^{-1+2\varepsilon},\ \|C\|_2 < e_k^{-1+3\varepsilon}\}.$$

The external-slot activity norm is precisely

$$\|E\|_k^x = \sup_X \exp(\kappa d_M(X)) \sup_{(A,C)\in \widetilde{R}_k\times\mathfrak{X}_k} \|E(X;A,C)\|_{h_k}.$$

For the no-slot family $E(X;A) = \mathscr{E}(X;A,0)$, write $\|E\|_k$ for the same supremum restricted to $C=0$. Assume the polymer counting bound $\sum_{X\supset\square} e^{-\kappa_0 d_M(X)} \le K_0$, $\kappa \ge \kappa_0$.

Here is the complete symmetry package. Use $\gamma_\mu = \sigma_3\otimes\sigma_\mu$, $\gamma_5 = \sigma_1\otimes 1_2$, $\mathsf{C} = i\gamma_2$, and $\varkappa = \gamma_5\mathsf{C}$. For a spin lift S generated by the four γ-matrices and a translation $a\in M\mathbb{Z}^3$, let $\widetilde{g}(c,g) = (r(S)c+a, r(S)g)$, where $S\gamma_\mu S^{-1} = r(S)_{\mu\mu}\gamma_\mu$. Define

$$a_{\widetilde{g}}\psi(x) = S\psi(\widetilde{g}^{-1}x), \qquad a_{\widetilde{g}}\bar{\psi}(x) = (S^{-1})^T\bar{\psi}(\widetilde{g}^{-1}x).$$

For pair generators transform both endpoints and use the same spin matrices. G_λ multiplies an unbarred ordinary or pair leg by $\exp(ie_k\lambda(c))$, with c its first photon point, and a barred leg by its inverse. Define on ordinary and pair legs

$$\mathrm{Cg}\,\psi = \mathsf{C}\bar{\psi}, \quad \mathrm{Cg}\,\bar{\psi} = -\mathsf{C}\psi, \qquad \mathbf{K}\psi = \varkappa\psi, \quad \mathbf{K}\bar{\psi} = \varkappa\bar{\psi},$$

with $\mathbf{K}$ antilinear. The complex activity class requires

$$\begin{aligned} a_{\widetilde{g}}E(\widetilde{g}X;\widetilde{g}A,\widetilde{g}C) &= E(X;A,C),\\ \mathrm{Cg}\,E(X;-A,-C) &= E(X;A,C),\\ G_{\lambda+\lambda'}E(X;A^1,C^1) &= E(X;A,C) \end{aligned}$$

whenever the two pairs are in the domain and $A^1 = A - d\lambda$, $C^1 = C - d\lambda'$ on bonds at distance at most one from X, for real λ,λ'. Its real subspace additionally requires

$$\mathbf{K}E(X;-\overline{A},-\overline{C}) = E(X;A,C).$$

All equations act on independent generators before evaluation. For the reconstruction application in RNO1, the explicit conditions

$$e_k^{\varepsilon} \le \tfrac{1}{4}, \qquad e_k^{1/4} + e_k^{\varepsilon} \le 1, \qquad 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \le 1$$

give $\widetilde{R}_k + \mathfrak{X}_k \subset \widetilde{G}_k$, where $\widetilde{G}_k$ has a global complex bound $e_k^{-1+\varepsilon}$ and local real-gauge sup bounds $e_k^{-1+\varepsilon}$ on every $11M$ cube. They also give $e_k\|\mathrm{Im}(A+C)\|_\infty \le 1$ and $e_k\|d(A+C)\|_\infty \le 1$.

Source. 263-page master Sections 12.1–12.2, TeX 10142–10370; the margin-reduced domain is used at 13164–13262. The activity notation $\mathcal{A}_k^{x,\mathbb{C}}$ here denotes the space $K_k^{x,\mathbb{C}}$ of 263-page master. This setup is shared with RNO1/D.